University of Michigan Survival Flight Clinical Guideline: B-1

Management of Adult Patients with Acute Alveolar Injury or Disease (ARDS)

Acute Respiratory Distress Syndrome (ARDS) is an injury to alveolar tissue resulting in refractory hypoxemia and carries a high mortality rate. These critically ill patients require aggressive, lifesaving therapies offered exclusively at the University of Michigan Medical Center and require transport by Survival Flight. This Clinical Guideline provides the evidence-based standard, mutually agreed upon by UMHS intensivists that will guide Survival Flight management during transport.

Required Communication with Accepting MD/ Fellow:

1. Prior to leaving base in order to establish management plan
2. Upon contact with patient / referring staff and initiation of therapies
3. Prior to leaving referring institution

- SICU Fellow ((734) 216-6974)
- CCMU Fellow (ICU (734) 936-4752 to page Fellow)
- UH Respiratory Care Supervisor (Pager 1550)

available for consultation as needed

Lung Protective Ventilation

- Vt 6 ml/kg Ideal (Predicted) Body Wt
- Pplat < 30 cmH_2O
  - Reduce Vt to 5 or 4 ml/kg if necessary

The Hamilton T1 takes several breaths in order to reach target Vt. For a patient in ARDS, this may de-escalate therapy. Consider initiation of Pressure Controlled Ventilation (PCV) for targeted existing Vt. PCV set as ΔP/PEEP

PRESSURE CONTROL (Starting Point)

(Goals are to match Minute Ventilation (Ve) and Mean Airway Pressure (MAP))

1. Match set ventilator rate
2. Target PIP for Exhaled Tidal Volume (Vte). (SHOULD BE CLOSE TO PREVIOUSLY ESTABLISHED PLATEAU PRESSURE (Pplat)
3. Match PEEP
4. Match Inspiratory Time
5. Titrate rate, Pressure Control (ΔP), Inspiratory time and PEEP for target Minute Ventilation (Ve) and MAP

Berlin Criteria for ARDS

TIMING within 1 week of a known clinical insult or new or worsening respiratory symptoms

CHEST IMAGING shows BIL opacities not fully explained by effusions, lobar/lung collapse or nodules

ORIGIN of EDEMA is respiratory failure not fully explained by cardiac failure or fluid overload. Objective assessment (e.g., echocardiography) to exclude hydrostatic edema if no risk factor present.

OXYGENATION PARAMETERS
Mild 200 mmHg < $PaO_2/FIO_2 \le 300$ mmHg with PEEP or CPAP ≥5 cmH_2O
Moderate 100 mmHg < $PaO_2/FIO_2 \le 200$ mmHg with PEEP ≥5 cmH_2O
Severe $PaO_2/FIO_2 \le 100$ mmHg with PEEP ≥5 cmH_2O

CONSULT MEDICAL CONTROL FOR:

- **SUSTAINED PIPs > 50 cmH_2O**
- **SUSTAINED PEEP > 20 cmH_2O**
- **ΔP > 25 cmH_2O**

Developed November 2009; Revised May 2024 (CRS/PMM/MJL)

Management Flowchart

Concurrent Medical Management **(Recruitment and CV Support Page)**

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ENSURE ADEQUATE SEDATION and NEUROMUSCULAR BLOCKADE

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Initiation of Ventilation with Hamilton T1 while attempting to establish **"Lung Protective Ventilation"** strategies[1,2,7]

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Adequate Oxygenation?[3]
- **YES** → Continue with current settings and package for transport
- **NO** → Switch to Pressure Control Ventilation **(PRESSURE CONTROL)** → Consider Initiation of Inhaled Nitric Oxide (iNO)[7] **10 ppm** → Stepwise Recruitment Maneuver **(Recruitment and CV Support Page)**

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Adequate Oxygenation?[3]
- **YES** → Continue with current settings and package for transport
- **NO** → Re-establish ventilation parameters with the minimum PEEP necessary to maintain target oxygenation goals[4]

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Adequate Oxygenation?[3]
- **YES** → Continue with current settings and package for transport
- **NO** → Lengthen I Time for <u>maximum</u> I:E ratio of 3:1[5]

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Critical AutoPEEP Present?[6]
- **YES** → Consider:
  - Slight Reduction in Inspiratory Time or Rate
  - Slight reduction in SET PEEP
  - Initiation / Titration of Pressors
  - Bronchodilators
- **NO** → back to Adequate Oxygenation?[3]

Concurrent Guidelines:

C-4.1: Cardiogenic Shock
C-4.2: Distributive Shock
S-1: Patient Comfort and Safety

NOTES

1. If in Volume Control, target expired, not set Tidal Volume (Vt). Ensure no air leak
2. Lung Protective Ventilation modified to ensure Mean Airway Pressure (MAP) and adequate oxygenation.
3. Oxygenation Target: SpO_2 / SaO_2 ≥88% and/or PaO_2 ≥60mmHg
4. Increase PEEP to maximum that patient will tolerate based upon blood pressure, heart rate and other hemodynamic parameters such as presence of peripheral / central pulses. Maximum PEEP that the Hamilton T1 will deliver is 30 cmH2O.
5. Maximum I:E for minimum acceptable hemodynamic and oxygenation parameters for transport. **DO NOT INVERSE I:E IN VOLUME VENTILATION!!**
6. The amount of AutoPEEP that would cause cardiovascular compromise secondary to excessive intrathoracic pressures
7. Ensure that the endotracheal tube is clamped during all circuit change evolutions (clamp for < 10 sec)
8. Pts that require special pathogen precautions will have viral filters attached to both the inspiratory and expiratory sides of the Hamilton T1 transport ventilator.

Oxygenation Target:

- SpO_2 / SaO_2 ≥88% and/or
- PaO_2 ≥60mmHg

Goal Transport Threshold

- pH ≥ 7.20
- PaO_2 60 mmHg

LUNG RECRUITMENT and CARDIOVASCULAR SUPPORT

Stepwise Recruitment Maneuver

1. Ensure patient is adequately sedated, paralyzed and in pressure control ventilation (PCV)
2. Increase PEEP **5 cmH_2O** above current setting. Peak Pressure will rise accordingly.
  - **MAXIMUM Pressure Control** (Δ P) setting prior to consulting with medical control is **25 cmH_2O**
3. Maintain increased PEEP for **1-3 minutes** (monitor for hemodynamic compromise) **IF NO IMPROVEMENT IN OXYGENATION THEN...**
4. Increase PEEP another 5 cmH_2O for 1-3 minutes (monitor for hemodynamic compromise).
5 .Repeat until a **maximum PEEP of 25 cmH_2O** is achieved for 2 minutes (monitor for hemodynamic compromise)
6. Recruitment will be terminated if one or more of the following signs of clinical deterioration are observed:
  - > 20% Δ in heart rate from baseline (or if HR drops lower than 60 bpm)
  - MAP drops below 60 mmHg
  - Decrease in SpO_2 below baseline for > 30 sec
7. Re-establish ventilation parameters with the minimum PEEP necessary to maintain target oxygenation goals. Attempt to maintain PIP< 35 cmH_2O
8. **CONTACT MEDICAL CONTROL FOR RECRUITMENT NEEDS ABOVE MAXIMUM SETTINGS NOTED ABOVE**

Suspicion of Clinical Deterioration Necessitating the need for Chest Tube Placement Secondary to Pneumothorax

1. Loss of breath sounds on one side of the chest
2. Sudden ventilator high pressure alarms
3. Sudden loss of tidal volumes (Vt)
4. Hemodynamic deterioration
5. Radiographic evidence of pathology

Concurrent Medical Management Goals

- Sedation / Analgesia / Neuromuscular Blockade per SF S-1 (Patient Comfort & Safety)
- Monitor / document CVP (or other preload parameters) and discuss status with **Medical Control for further guidance with respect to need for fluid resuscitation vs. diuresis**. (LATE SEPSIS and/or SHOCK WITH HIGH SVR STATES MAY BENEFIT FROM DIURESIS).
- MAP $\geq$ 65 mmHg
  - Norepinephrine (0.01-1 mcg/kg/min) and vasopressin (0.04 units/min) to keep MAP $\geq$ 65 mmHg
  - Consider inotropic support (milrinone, dobutamine or epinephrine) to enhance RV contractility and promote improved pulmonary perfusion (Refer to *Cardiovascular Support* Table)
- Arterial pH $\geq$ 7.2, (**Contact Medical Control for Guidance)**
  - If < 7.2 and PaCO2 < 60, consider Sodium Bicarbonate. Serum Sodium Limit 155 mEq/L
  - If < 7.2 and PaCO2 > 60 and/ or elevated serum sodium, consider **THAM** per **Figure 1**
- Ionized Calcium $\geq$ 1.2 mmol/L (OR $\geq$ 4.8 mg/dl) (Normal Ionized Calcium levels: 1.12-1.3 **mmol/L** OR 4.48-5.2 **mg/dl**)
  - If low and patient is hypotensive, consider Calcium Chloride 1 GM / dose via a central line OR Calcium Gluconate 1-2 GM / dose
  - Reassess iCa+ with initiation / continuation of therapy

Cardiovascular Support							
Agent	**Dose (Std UMHS Conc.)**	**Mechanism**	**HR**	**Systolic Function**	**SVR**	**PVR**	**Comments**
Dobutamine	**5-10 mcg/kg/min (1000 mg / 250 ml) PREMIX**	**Pure β; anti-α**	**Increase**	**Increase**	**Decrease**	**Minimal Decrease**	**Consider for increased RV contractility, MAP > 65 mmHg and $ScvO_2$< 70%**
Epinephrine	**0.01-1 mcg/kg/min (5 mg/250 ml)**	**B_1 agonist > α agonist**	**Increase**	**Significant Increase**	**Increase**	**Minimal Increase**	**Consider for increased RV contractility, MAP < 65 mmHg**
Norepinephrine	**0.01-1 mcg/kg/min (16 mg / 250 ml)**	**B_1 agonist < α agonist**	**Some Increase**	**Some Increase**	**Significant Increase**	**Minimal Increase**	**Drug of Choice for hypotension with low SVR**
Milrinone	**0.125-0.75 mcg/kg/min (20 mg / 100 ml) PREMIX**	**Phosphodiesterase Inhibitor**	**No change**	**Increase**	**Decrease**	**Decrease**	**Consider for increased RV contractility, MAP > 65 mmHg and $ScvO_2$< 70% (can drop BP more than dobutamine)**
Vasopressin	**0.04 units/ min (100 units / 100 ml)** ***2.4 ml/hr @ this conc./dose***	**ADH V Receptor Agonist**	**No Effect**	**No Effect**	**Significant increase**	**Unknown Effect**	**SNS agent sparing. Use in conjunction with norepinephrine**

PREDICTED (IDEAL) BODY WEIGHT (PDW)

FEMALES

HEIGHT	PBW	4 ml	5 ml	6 ml	7 ml	8 ml
4' 0" (48)	17.9	72	90	107	125	143
4' 1" (49)	20.2	81	101	121	141	162
4' 2" (50)	22.5	90	113	135	158	180
4' 3" (51)	24.8	99	124	149	174	198
4' 4" (52)	27.1	108	136	163	190	217
4' 5" (53)	29.4	118	147	176	206	235
4' 6" (54)	31.7	127	159	190	222	254
4' 7" (55)	34	136	170	204	238	272
4' 8" (56)	36.3	145	182	218	254	290
4' 9" (57)	38.6	154	193	232	270	309
4' 10" (58)	40.9	164	205	245	286	327
4' 11" (59)	43.2	173	216	259	302	346
5' 0" (60)	45.5	182	228	273	319	364
5' 1" (61)	47.8	191	239	287	335	382
5' 2" (62)	50.1	200	251	301	351	401
5' 3" (63)	52.4	210	262	314	367	419
5' 4" (64)	54.7	219	274	328	383	438
5' 5" (65)	57	228	285	342	399	456
5' 6" (66)	59.3	237	297	356	415	474
5' 7" (67)	61.6	246	308	370	431	493
5' 8" (68)	63.9	256	320	383	447	511
5' 9" (69)	66.2	265	331	397	463	530
5' 10" (70)	68.5	274	343	411	480	548
5' 11" (71)	70.8	283	354	425	496	566
6' 0" (72)	73.1	292	366	439	512	585
6' 1" (73)	75.4	302	377	452	528	603
6' 2" (74)	77.7	311	389	466	544	622
6' 3" (75)	80	320	400	480	560	640
6' 4" (76)	82.3	329	412	494	576	658
6' 5" (77)	84.6	338	423	508	592	677
6' 6" (78)	86.9	348	435	521	608	695
6' 7" (79)	89.2	357	446	535	624	714
6' 8" (80)	91.5	366	458	549	641	732
6' 9" (81)	93.8	375	469	563	657	750
6' 10" (82)	96.1	384	481	577	673	769
6' 11" (83)	98.4	394	492	590	689	787
7' 0" (84)	100.7	403	504	604	705	806

MALES

HEIGHT	PBW	4 ml	5 ml	6 ml	7 ml	8 ml
4' 0" (48)	22.4	90	112	134	157	179
4' 1" (49)	24.7	99	124	148	173	198
4' 2" (50)	27	108	135	162	189	216
4' 3" (51)	29.3	117	147	176	205	234
4' 4" (52)	31.6	126	158	190	221	253
4' 5" (53)	33.9	136	170	203	237	271
4' 6" (54)	36.2	145	181	217	253	290
4' 7" (55)	38.5	154	193	231	270	308
4' 8" (56)	40.8	163	204	245	286	326
4' 9" (57)	43.1	172	216	259	302	345
4' 10" (58)	45.4	182	227	272	318	363
4' 11" (59)	47.7	191	239	286	334	382
5' 0" (60)	50	200	250	300	350	400
5' 1" (61)	52.3	209	262	314	366	418
5' 2" (62)	54.6	218	273	328	382	437
5' 3" (63)	56.9	228	285	341	398	455
5' 4" (64)	59.2	237	296	355	414	474
5' 5" (65)	61.5	246	308	369	431	492
5' 6" (66)	63.8	255	319	383	447	510
5' 7" (67)	66.1	264	331	397	463	529
5' 8" (68)	68.4	274	342	410	479	547
5' 9" (69)	70.7	283	354	424	495	566
5' 10" (70)	73	292	365	438	511	584
5' 11" (71)	75.3	301	377	452	527	602
6' 0" (72)	77.6	310	388	466	543	621
6' 1" (73)	79.9	320	400	479	559	639
6' 2" (74)	82.2	329	411	493	575	658
6' 3" (75)	84.5	338	423	507	592	676
6' 4" (76)	86.8	347	434	521	608	694
6' 5" (77)	89.1	356	446	535	624	713
6' 6" (78)	91.4	366	457	548	640	731
6' 7" (79)	93.7	375	469	562	656	750
6' 8" (80)	96	384	480	576	672	768
6' 9" (81)	98.3	393	492	590	688	786
6' 10" (82)	100.6	402	503	604	704	805
6' 11" (83)	102.9	412	515	617	720	823
7' 0" (84)	105.2	421	526	631	736	842

ARDSNet

Figure 6: Tromethamine (THAM 0.3M Solution)

Contraindications: Uremia or anuria.
- Specific contraindications in neonates: Chronic respiratory acidosis or Salicylate intoxication

Adverse reactions:
- Hypoglycemia: may cause persistent hypoglycemia with rapid administration or large doses
- Fluid overload: may cause fluid &/or solute overload resulting in dilution of serum electrolyte concentrations, overhydration, congested states or pulmonary edema
- Hyperkalemia: monitor for s/sx of hyperkalemia in pts with renal failure or decreased urine output due to decreased THAM excretion (via kidneys)
- Apnea/Respiratory depression: large doses may depress ventilation.
- Vesicant: Central line preferred. Ensure proper IV placement prior to & during THAM administration.
  (Neonates: Avoid infusion via low-lying umbilical venous catheters due to risk of hepatocellular necrosis)

Dosing calculation:

Required **millileters (mL)** of THAM solution per dose = **actual body weight (kg) x base deficit (mEq/L) x 1.1**

(e.g.: 90 kg pt with a base deficit ("negative base excess") of 5 mEq/L would require 90 x 5 x 1.1 = 495 ml)

Administration: Avoid rapid administration. When feasible, infuse over 1 hour.
Utilize largest IV available, central line preferred.

Monitoring:
- Assess for signs of fluid overload prior to and during administration
- Assess blood glucose & pH prior to initiation/redosing and 30 mins after completion.
- Monitor for signs of Hyperkalemia in pts with renal dysfunction or decreased urine output.